Tuning and Delay-Aware State Prediction in Cart-Pole Survival Boundaries

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Abstract—Controller rankings near a delay-induced failure cliff can reflect filter tuning, noise units and stale-data handling as much as feedback architecture. We study point-mass and uniform-rod cart poles with equal mass and centre-of-mass distance. A paired nonlinear benchmark corrects the continuous observer covariance units, sweeps finite-difference filtering with fixed and redesigned gains, and extends process-to-measurement covariance ratios across fourteen orders of magnitude. Tunings are selected on independent development trials. A timestamp-aware Kalman predictor is compared with filters that deliberately treat delayed measurements as current. Held-out batches of 500 trials estimate the first observed downward crossings of 50% ten-second survival on the evaluated delay grid, with 1 ms local brackets for 5 N and 10 N pushes. For the point model, the reference finite-difference first observed crossing is 51.9 ms at 5 N, while filter-aware redesign reaches 60.7 ms. The base discrete filter first observed crossings are 66.3 and 44.1 ms; delay-aware prediction gives 252.2 and 125.5 ms. Phase margin divided by crossover closely predicts nominal sampled-loop instability but is tested separately against nonlinear survival. Survival, terminal dwell and active-time effort expose distinct noise-versus-lag trade-offs. The contribution is a calibrated and reproducible boundary comparison, not a new estimator or a general controller hierarchy.

Index Terms—Cart pole, delayed measurements, Kalman prediction, baseline calibration, sampled control, survival boundary.

I INTRODUCTION

The inverted pendulum remains useful for studying unstable mechanical systems and constrained feedback [1]. Its simplicity also makes controller comparisons sensitive to seemingly minor implementation choices. A derivative filter changes loop dynamics, a covariance ratio changes estimator bandwidth, and a delayed measurement requires a timestamp-consistent update. A favorable success rate at one delay does not isolate any one of these mechanisms.

Comparisons of LQG-oriented pendulum control [2] and broader control families [3] already exist. Reproducible benchmark environments support disturbances in inputs, measurements and inertia [4], [5]. Delayed-measurement Kalman filtering and state augmentation are established methods [6], and the absence of general LQG margin guarantees is classical [7]. Consequently, neither a PID/LQR/LQG comparison, a predictor, nor the observation that a filtered loop has different margins constitutes algorithmic novelty.

The question addressed here is narrower than proposing another pendulum controller: for a specified task and information set, which apparent delay advantages persist after correcting noise units and sample timing, calibrating derivative filters and gains, and using known measurement age? Existing comparative studies [2], [3] assess controller performance, benchmark suites [4], [5] provide repeatable environments, and delayed-estimation theory [6] provides the predictor. We do not claim a missing general benchmark or new predictor theory. Our empirical contribution is the controlled combination of these checks in two matched inertia models, with trial-paired first observed crossings, their recrossings, and direct comparisons with nominal linear delay predictions. This is a case-specific attribution study, not evidence that the combination is absent from all earlier literature.

II MODEL AND TASK

The state is $z = [x, v, \theta, \omega]^T$, with θ measured from upright and positive toward positive cart displacement. Cart mass $M = 1$ kg, pole mass $m = 0.1$ kg, centre-of-mass distance $l = 0.5$ m, viscous cart friction $b = 0.1$ N s/m and $g = 9.81$ m/s² define the nominal plants. Let $I = \rho m l^2$ and $J = I + ml^2$. The point model uses $\rho = 0$; the uniform rod uses $\rho = 1/3$ and total length $2l$. Thus the two plants share COM distance rather than tip length.

$$M_q(\theta)\begin{bmatrix}\ddot{x}\\ \ddot{\theta}\end{bmatrix} = \begin{bmatrix}u + f_{\text{push}} - bv + ml\omega^2\sin\theta\\ mgl\sin\theta\end{bmatrix} \quad (1)$$
$$M_q(\theta) = \begin{bmatrix}M+m & ml\cos\theta\\ ml\cos\theta & J\end{bmatrix}$$

Equation (1) gives the complete coupled dynamics, including the clipped command u, external push fpush, cart friction and centrifugal term. Here Mq is the generalized mass matrix and $J = I + ml^2$. Its determinant is positive for the declared positive masses. The attached supplement derives both accelerations, energy balance, linearization and exact sampled transitions. Gains are designed for each nominal inertia model.

The nonlinear plant uses fourth-order Runge–Kutta integration at 1 ms and a 10 ms control period with zero-order-held commands clipped to ±10 N. The initial state is $[0,0,0.1,0]^T$. An unannounced external push of 5 N or 10 N acts from 2.0 through 2.1 s, after command clipping. Each trial independently draws M, m, l and b uniformly within ±20% of nominal and holds them fixed. Position and angle measurements have independent per-sample standard deviations 0.005 m and 0.002 rad. No Gaussian process noise is injected into the plant; W is an estimator design covariance for unmodelled dynamics.

$$y_k = Cz(t_k - \tau) + v_k \quad (2)$$

The delay τ is constant and known only to the predictor. Measurements refer to the actual 1 ms plant history,

permitting fractional control-sample delays. Noise is generated at 100 Hz for each arriving packet. No packet predates initialization: controllers hold the known initial packet until a valid delayed packet arrives. The estimator initial mean is the stated initial state; the predictor uses a conservative steady-state prior covariance. All other filters intentionally ignore packet age.

A trial survives if $|\theta| \le 0.5$ rad and $|x| \le 2.4$ m throughout ten seconds. Failure freezes the state immediately and sets commands to zero at the next scheduled control update. Terminal dwell additionally requires $|\theta| \le 0.002$ rad and $|x| \le 0.02$ m continuously over the final second. Active-time RMS force divides effort by time before termination; full-horizon RMS is supplied separately. Neither low effort after failure nor finite-horizon survival is described as successful stabilization.

III BASELINES AND CALIBRATION

All LQR designs use Q = diag(1,1,25,1), R = 0.01 and $u = -K\hat{z}$. The continuous regulator uses CARE. The discrete regulator uses exact ZOH dynamics and stage weights QTs and RTs, a stated sampled-stage approximation rather than an exact intersample cost. The velocity oracle transports exact velocities with noisy positions and is an informational reference, not a position-only implementation.

$$H_v(z) = \frac{(1-\alpha)(1-z^{-1})}{T_s(1-\alpha z^{-1})} \quad (3)$$

The finite-difference reference uses α = 0.8. We sweep $\alpha \in \{0,0.1,0.2,0.4,0.6,0.8,0.9,0.95,0.98\}$, first keeping continuous LQR gains fixed. A second sweep redesigns the four gains while including filter memory in the closed loop. The local static-output-feedback objective combines the nominal initial-condition quadratic transient and a ten-second stationary-noise H_2 surrogate. It is solved from multiple starts with bounded gain multipliers; it is not a global or delay-robust optimum. PD reference and LQR-FD are the same four-gain filtered feedback family after their gains are specified, so their labels are not treated as independent architecture classes.

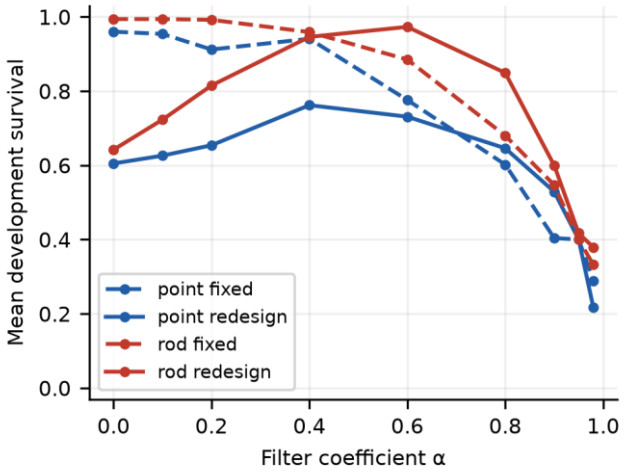


Fig. 1. Development sweep over nine finite-difference filter coefficients, with fixed continuous LQR gains and a local filter-aware gain redesign. Each point averages five declared tasks and 200 paired trials per task. Tuning uses development data only; no global optimality is asserted.

Continuous Kalman design requires a measurement spectral density rather than per-sample variance. Under the equivalent independently sampled/boxcar-average white-noise convention we use

$$V_c = R_m T_s \quad (4)$$

At Ts = 0.01 s, inserting Rm directly gives a spectral density 100 times too large. Multiplying W by 100 with the legacy V produces the same gain as correcting V, to a verified relative error below 3×10^{-12}. This is a covariance-ratio intervention and cannot be attributed solely to observer implementation. Continuous and discrete filtering are still different sampled algorithms, so the conversion alone does not make their gains identical.

We retain a legacy-current-sample update solely to isolate the unit correction: both its old-unit and corrected-unit versions integrate the newly received sample over the preceding interval before computing control. This timing is not presented as a causal continuous-observer implementation. The causal version instead propagates with the previous held sample and previous clipped command, then holds the newly received sample for the next interval. Discrete filters predict with the known clipped command and then update, using exact Qd [8] and Rm.

The corrected causal continuous observer and discrete filter are each swept over Wσ = σW for $\sigma = 10^j$, j = −6,…,8, with fixed feedback gains. The Kalman gain depends on the ratio W/V, not their common magnitude. Selection maximizes mean survival across the five development tasks (0 ms/5 N, 40 ms/5 N, 60 ms/5 N, 0 ms/10 N, 60 ms/10 N), then mean dwell and finally minimizes active RMS. Each setting uses 200 trial-paired runs per task. The same declared task weights apply to each family. Selected tunings are frozen before boundary estimation; all sweep points remain available.

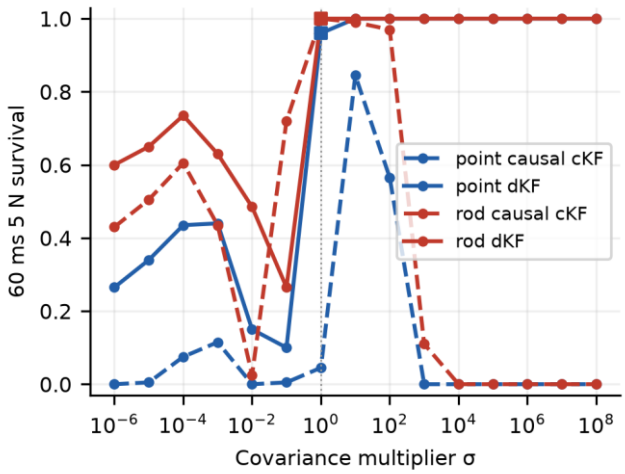


Fig. 2. Development survival at 60 ms/5 N over $\sigma = 10^{-6}$ through 10^8. Squares mark dKF at σ = 1: point dKF 192/200 (96.0%) versus held-out 485/500 (97.0%); rod dKF 200/200 (100.0%) versus held-out 499/500 (99.8%). Fig. 2 uses seed 20261121 and 200 trials per point; Table II/Fig. 6 use seed 20261122 and 500. Solid curves are dKF; dashed curves are causal cKF. The selection objective averages all five declared tasks.

The selected fixed-gain α values are 0.0 and 0.1 for point and rod, respectively. Filter-aware redesign selects 0.4 and

0.6. The selected continuous-observer multipliers are 10 and 1; discrete-filter multipliers are 10 and 100. These are best tested development settings under the declared lexicographic objective. Decade-spaced maxima and high-end plateaus do not identify a unique continuous optimum or a universal performance advantage. The supplement gives the complete dwell and effort sweeps.

IV KNOWN DELAY PREDICTION

The delay-unaware discrete design fuses stale measurements as current and is therefore a robustness baseline. The predictor instead maintains an estimate at the measurement timestamp, performs the Kalman innovation there, and propagates it to the current control time with recorded clipped commands. For $\tau = qTs$,

$$\hat{z}_k = F^q \hat{z}_{k-q|k} + \sum_{h=0}^{q-1} F^h G u_{k-1-h} \quad (5)$$

For $\tau = qTs + r$, $0 \le r < Ts$, exact fractional transitions integrate the partial held-input intervals; the supplement provides the full expression. Unknown external push values and perturbed plant parameters are never supplied to the predictor. We evaluate both the base covariance and the multiplier selected for the delay-unaware discrete filter, without retuning the predictor on held-out data. This is an established known-delay baseline [6], not a proposed new filtering method. At zero delay it matches the ordinary discrete filter exactly. With the nominal linear model, noise-free measurements and no unknown push, it reproduces the zero-delay trajectory for tested delays up to 115 ms, with maximum error below 10^{-8}.

V DELAY BOUNDARIES AND UNCERTAINTY

$$D_{50} = \inf\{\tau \ge 0 : p_T(\tau) < \tfrac{1}{2}\} \quad (6)$$

Here pT is the ten-second survival probability at the fixed push magnitude. A separate 100-trial locator seed evaluates 0,20,40,60,80,100,150,200 and 300 ms. Held-out batches use 500 new trials, verify and expand the bracket if needed, and bisect to a 1 ms crossing interval. A local 1 ms grid around the crossing checks recrossings. The initial locator ends at 300 ms, but brackets may expand up to an administrative search ceiling of 1200 ms. The largest delays actually evaluated were 300 ms for the point model and 600 ms for the rod; rod-predictor brackets expanded from 300 to 600 ms before refinement. All 56 reported crossings are interior, so none is right-censored. Interpolated D50 values and 95% intervals use 2,000 paired resamples of trial identities, preserving each trial's uncertainty and noise sequence across delays. Intervals are pointwise and conditional on the evaluated adaptive grid; they do not incorporate a repeated tuning search, certify monotonic survival, or provide simultaneous coverage. Coarse locator and local-grid recrossings are audited explicitly.

The development, locator and held-out seeds are distinct. Common random numbers are invariant to case ordering and batch size. A first observed downward half-survival crossing is reported; below-half survival at zero delay and failure to cross by the tested limit would be censored, not assigned a fictitious optimum. Fixed 0,40 and 60 ms cells support paired intervention effects with bootstrap intervals and exact McNemar tests; Holm correction covers all 108 declared comparisons [9], [10], [11].

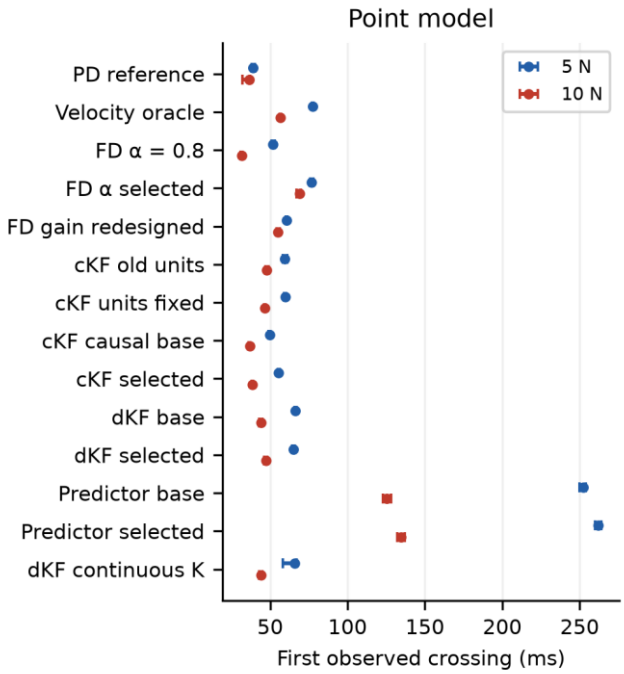


Fig. 3. Held-out point model first observed downward crossings of 50% ten-second survival for 5 N and 10 N pushes. Bars are 95% trial-paired bootstrap intervals conditional on evaluated delays; local numerical crossing brackets are 1 ms. A cross at zero denotes survival below 50% without added delay. These are not monotone delay limits or stability certificates.

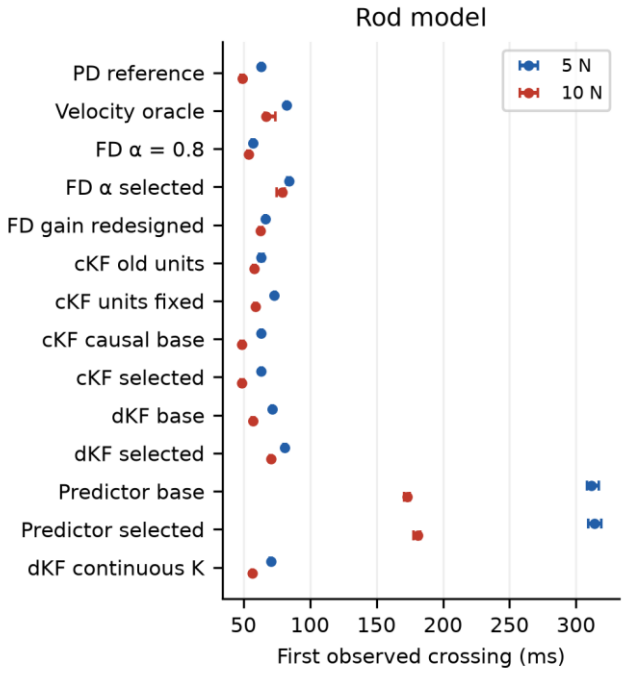


Fig. 4. Held-out rod model first observed downward crossings of 50% ten-second survival for 5 N and 10 N pushes. Bars are 95% trial-paired bootstrap intervals conditional on evaluated delays; local numerical crossing brackets are 1 ms. A cross at zero denotes survival below 50% without added delay. These are not monotone delay limits or stability certificates.

Table I compares designs tuned on tasks that include 40 and 60 ms delays, near some of the crossings measured here. New Monte Carlo seeds separate trial realizations; this is not an out-of-delay-distribution tuning test. The predictor reuses the delay-unaware discrete filter's selected covariance ratio. Four-gain derivative redesign and scalar Kalman-ratio sweeps have unequal tuning budgets, so these are declared calibrated designs rather than equally optimized controller classes.

TABLE I FIRST OBSERVED 50 PERCENT SURVIVAL CROSSINGS IN MS

Feedback	Point 5 N	Point 10 N	Rod 5 N	Rod 10 N
PD reference	38.9	36.3	63.3	49.0
Velocity oracle	77.4	56.6	82.6	66.8
FD reference	51.9	31.7	57.3	54.1
FD selected	76.7	69.0	84.1	79.2
FD redesigned	60.7	55.1	66.5	63.0
cKF legacy	59.2	47.8	63.4	58.2
cKF units fixed	59.6	46.7	73.2	59.0
cKF causal	49.6	36.7	63.1	48.9
cKF selected	55.5	38.6	63.1	48.9
dKF base	66.3	44.1	71.5	56.9
dKF selected	64.9	47.3	81.2	70.5
Predictor base	252.2	125.5	311.4	173.3
Predictor selected	262.2	134.5	314.0	181.1
dKF continuous K	65.9	44.1	70.9	56.8

For point-mass base dLQG, D50 changes from 66.3 [66.0, 67.0] ms at 5 N to 44.1 [43.0, 45.0] ms at 10 N. The corresponding base-predictor values are 252.2 [250.0, 254.0] and 125.5 [123.0, 128.0] ms. Thus a single 60 ms success rate cannot stand for delay robustness. The tuned filter and filter-aware derivative baseline also alter the ranking; boundary and tuning information are necessary to interpret individual success cells.

VI MARGINS AND NONLINEAR RESULTS

For each fixed compensator we compute all nominal zero-delay sampled gain crossovers, retaining positive phase margins φmi in radians. A pure added-delay estimate is

$$\tau_{\mathrm{PM}} = \min_i \frac{\varphi_{\mathrm{mi}}}{\omega_{\mathrm{gci}}} \quad (7)$$

We independently form the exact fractional-measurement-delay closed loop by augmenting plant history, held command history and controller state, and bracket the first observed spectral-radius crossing at one millisecond. This validates the linear calculation against the actual sampled implementation, rather than borrowing a full-state LQR margin. A predictor changes with τ, so the fixed-loop division is not applied to it. In its ideal known-model setting, prediction removes deterministic measurement age from nominal regulation; unknown disturbances and model errors still limit the nonlinear task.

The point PD and legacy-observer phase margins are 27.3° and 27.8°, yet their crossover frequencies are 13.66 and 8.65 rad/s. The resulting estimates are 34.9 and 56.2 ms, compared with exact lifted boundaries of 34.5 and 56.5 ms. Similar phase margins therefore do not imply similar delay margins. Across fixed loops, the maximum discrepancy between the phase-based estimate and the midpoint of the one-millisecond lifted bracket is 0.5 ms.

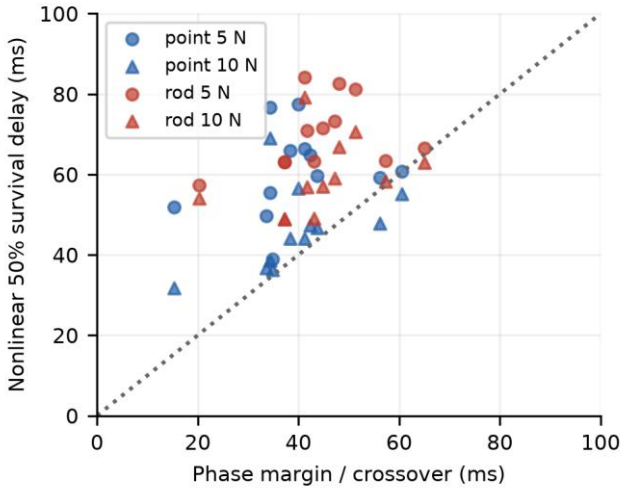


Fig. 5. Phase-based nominal delay estimates versus nonlinear ten-second survival boundaries. The diagonal is equality. Exact lifted linear predictions, tabulated in the supplement, closely match the phase-based values. Predictors are omitted because their compensators depend on delay.

Figure 5 tests, rather than assumes, the link to nonlinear survival. Linear loss of local asymptotic stability, bounded saturated oscillation and finite-horizon failure are different events. Parameter perturbations change poles and crossings; pushes excite nonlinear excursions; command clipping can either prolong bounded survival or prevent recovery. Consequently, mismatch with D50 is an empirical result and not evidence that phase margin has failed as a local linear metric. We do not claim that margins alone explain the nonlinear ranking, and do not present the established LQG-margin limitation [7] as a novel result.

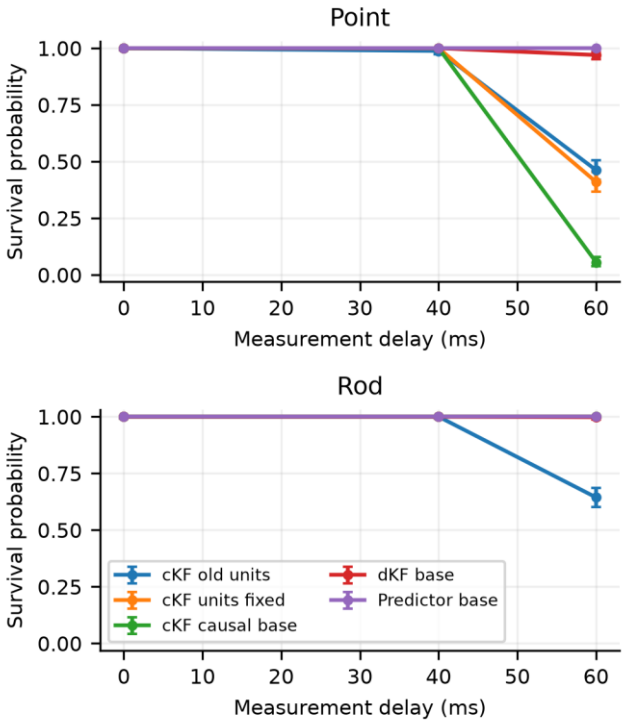


Fig. 6. Paired held-out checks isolate the covariance-unit correction within the legacy update, then enforce causal observer timing and compare delay-unaware and timestamp-aware discrete estimators. Each cell has 500 trials and pointwise Wilson intervals.

At point-model 60 ms/5 N, correcting units alone changes survival from 46.2% to 41.0%. The causal corrected continuous observer gives 5.6%, while a discrete filter retaining the same continuous regulator gain gives 92.4% and the discrete regulator gives 97.0%. Thus the unit error is real, but a unit correction alone does not recover the discrete-filter result in this task. The diagnostic with continuous K separates regulator discretization from the estimator update; neither change is treated as a universal implementation advantage.

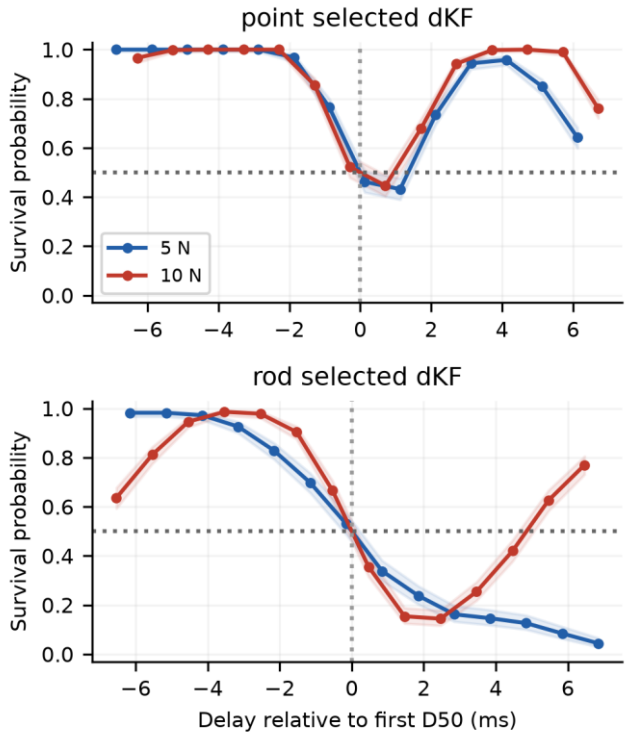


Fig. 7. Local one-millisecond held-out survival curves around each first observed half-survival crossing. Bands are pointwise Wilson intervals. Recrossings are retained: a single monotone collapse is not assumed, and reported boundary intervals condition on the evaluated delay grid.

The local curves in Fig. 7 retain upward half-survival recrossings. These can arise from the interaction of sampled measurement age, saturated trajectories and finite-horizon criteria; their presence prevents a claim of a unique monotone failure cliff. The full queried grid and recrossing counts are therefore part of the result, and D50 consistently denotes the first observed crossing.

TABLE II POINT MODEL NOISE LAG AND EFFORT TRADE OFF

Feedback	0 ms dwell	0 ms RMS N	60 ms survival
FD reference	0.0%	2.83	0.8%
FD selected	0.0%	8.14	100.0%
FD redesigned	0.4%	1.13	65.0%
cKF legacy	33.2%	0.87	46.2%
cKF causal	0.0%	1.19	5.6%
cKF selected	0.0%	2.05	87.0%
dKF base	0.6%	1.05	97.0%
dKF selected	0.0%	1.78	100.0%
Predictor base	0.6%	1.05	100.0%

Table II and the full sweeps show why survival alone is an incomplete objective. Low-bandwidth observation can improve quiet regulation and reduce force noise while increasing lag; aggressive observation or unfiltered differences can improve a delayed survival cell while increasing effort and destroying the strict terminal dwell event. Values on this frontier are not evidence for an inherent continuous-versus-discrete or derivative-versus-Kalman hierarchy. The velocity oracle is explicitly excluded from deployable rankings because it has additional information.

A paired onset-phase check shifts the push by an independent uniformly drawn integer from 0 through 9 ms while keeping its 100 ms duration and the same parameter and noise realizations. Across 269 cells, the largest observed absolute survival change is 2.8 percentage points at 60 ms and 5.4 points at queried local crossing anchors; 7 cells change sides of 50%. Point-model base dKF at 60 ms/5 N changes from 97.0% to 94.2% (paired difference -2.8 points, pointwise 95% interval -4.2 to -1.6). Supplement S1 gives the protocol, Fig. S5 and full tables. This check does not estimate a complete phase-averaged delay curve or prove monotone survival.

VII LIMITATIONS AND CONCLUSION

Cross-model differences include model-specific gains and therefore do not isolate inertia with a fixed controller. The primary task fixes push onset and duration. The added onset-phase check varies only the position of that push within one control period; it does not cover arbitrary disturbance times or randomized durations, and does not identify every cause of survival recrossing.

These results are simulation evidence for one plant scale, initialization, uncertainty law, measurement-noise level, command limit and ten-second horizon. Process covariance is a tuning device rather than an identified physical-noise spectrum. Filter-aware redesign is local and nominal; neither its gain search nor the predictor covariance is claimed globally optimal. Delay is fixed and exactly timestamped; jitter, packet loss, unknown initial conditions, sensor bias, friction nonlinearities and actuator dynamics require additional evaluation. Confidence intervals quantify trial variability under the declared distribution, not modelling uncertainty or hardware reliability.

The contribution is the controlled boundary study and its reproducibility: covariance-unit correction, causal timing, filter and gain calibration, known-delay prediction and a tested link between linear margins and nonlinear survival. Noise units, tuning objectives and packet timestamps must accompany task-specific controller comparisons. The results support an empirical methods and benchmarking study; they do not establish a new control method or a general controller hierarchy.

Data availability—Supplementary Material S1, Detailed Derivations and Reproducibility Supplement, is attached as a separate PDF and editable Word file. The versioned software, data, scientific figures and illustrative movies are archived with the repository release [12]. Each movie is a single paired realization, not a Monte Carlo success estimate.

Project repository | Archive DOI 10.5281/zenodo.24017779 | Supplement S1 PDF

REFERENCES

[1] K. J. Åström and R. M. Murray, Feedback Systems: An Introduction for Scientists and Engineers. Princeton University Press, 2008. https://authors.library.caltech.edu/records/yzs24-xsx88

[2] M. Rani and S. S. Kamlu, “Optimal LQG controller design for inverted pendulum systems using a comprehensive approach,” Scientific Reports, vol. 15, art. 4692, 2025, doi: 10.1038/s41598-025-85581-3.

[3] S. Irfan, L. Zhao, S. Ullah, A. Mehmood, and M. F. U. Butt, “Control strategies for inverted pendulum: A comparative analysis of linear, nonlinear, and artificial intelligence approaches,” PLOS ONE, vol. 19, no. 3, e0298093, 2024, doi: 10.1371/journal.pone.0298093.

[4] Z. Yuan et al., “safe-control-gym: A unified benchmark suite for safe learning-based control and reinforcement learning in robotics,” arXiv:2109.06325v4, 2022. https://arxiv.org/abs/2109.06325

[5] Y. Duan, X. Chen, R. Houthooft, J. Schulman, and P. Abbeel, “Benchmarking deep reinforcement learning for continuous control,” Proc. ICML, PMLR, vol. 48, pp. 1329–1338, 2016. https://proceedings.mlr.press/v48/duan16.html

[6] T. D. Larsen, N. A. Andersen, O. Ravn, and N. K. Poulsen, “Incorporation of time delayed measurements in a discrete-time Kalman filter,” Proc. 37th IEEE Conf. Decision and Control, vol. 4, pp. 3972–3977, 1998, doi: 10.1109/CDC.1998.761918.

[7] J. C. Doyle, “Guaranteed margins for LQG regulators,” IEEE Trans. Autom. Control, vol. 23, no. 4, pp. 756–757, 1978, doi: 10.1109/TAC.1978.1101812.

[8] C. F. Van Loan, “Computing integrals involving the matrix exponential,” IEEE Trans. Autom. Control, vol. 23, no. 3, pp. 395–404, 1978, doi: 10.1109/TAC.1978.1101743.

[9] E. B. Wilson, “Probable inference, the law of succession, and statistical inference,” J. Amer. Stat. Assoc., vol. 22, no. 158, pp. 209–212, 1927, doi: 10.1080/01621459.1927.10502953.

[10] Q. McNemar, “Note on the sampling error of the difference between correlated proportions or percentages,” Psychometrika, vol. 12, no. 2, pp. 153–157, 1947. https://pubmed.ncbi.nlm.nih.gov/20254758/

[11] S. Holm, “A simple sequentially rejective multiple test procedure,” Scandinavian J. Stat., vol. 6, no. 2, pp. 65–70, 1979. https://sci2s.ugr.es/keel/pdf/specific/articulo/0052_001.pdf

[12] R. M. Barathkumar, “Tuning and Delay-Aware State Prediction in Cart-Pole Survival Boundaries”, software and data, ver. 1.0.0, Zenodo, 2026, doi: 10.5281/zenodo.24017779. Repository: https://github.com/Ranger915/Cart-pole-research-project.

Detailed Derivations and Reproducibility Supplement

RAAGHAVAN MOOKIAH BARATHKUMAR

This supplement develops the mechanical model, regulator and estimator equations, filter-aware gain redesign, timestamp-aware prediction, exact fractional-delay loop and statistical procedure used in the accompanying article. Classical mechanical and quadratic-control notation follows [1]. All displayed equations are editable Word equations. The supplied numerical data retain the declared development/held-out separation.

1 Coordinates energies and equations of motion

Let cart position be x and pole angle from upward vertical be θ. The pole COM is (X,Y) = (x + l sinθ, l cosθ). Differentiation gives $\dot{X} = v + l\omega\cos\theta$ and $\dot{Y} = -l\omega\sin\theta$. Squaring and adding eliminates the sine/cosine square terms:

$$X = l\sin(\theta) + x \quad \text{(S1)}$$

$$Y = l\cos(\theta) \quad \text{(S2)}$$

$$V_{\text{squared}} = l^2\omega^2 + 2l\omega v\cos(\theta) + v^2 \quad \text{(S3)}$$

$$T = lm\omega v\cos(\theta) + \frac{\omega^2(I+l^2m)}{2} + \frac{v^2(M+m)}{2} \quad \text{(S4)}$$

$$V = glm\cos(\theta) \quad \text{(S5)}$$

$$I = 0 \quad \text{(S6)}$$

$$I = \frac{l^2m}{3} \quad \text{(S7)}$$

The generalized forces are $Qx = u + f_{push} - bv$ and $Q\theta = 0$. Applying $d/dt(\partial T/\partial\dot{q}) - \partial T/\partial q + \partial V/\partial q = Q$ yields $(M+m)\ddot{x} + ml\cos\theta\,\ddot{\theta} - ml\sin\theta\,\omega^2 = u + f_{push} - bv$ and $ml\cos\theta\,\ddot{x} + J\ddot{\theta} = mgl\sin\theta$. The cross terms in the θ equation cancel exactly. The upright equilibrium is a potential-energy maximum, so the linearized angular gravitational coefficient is positive.

$$\begin{bmatrix} M+m & lm\cos(\theta) \\ lm\cos(\theta) & I+l^2m \end{bmatrix} \quad \text{(S8)}$$

$$\Delta = -l^2m^2\cos(\theta)^2 + (I+l^2m)(M+m) \quad \text{(S9)}$$

$$a = \frac{-gl^2m^2\sin(\theta)\cos(\theta) + (I+l^2m)(-bv+lm\omega^2\sin(\theta)+u)}{-l^2m^2\cos(\theta)^2 + (I+l^2m)(M+m)} \quad \text{(S10)}$$

$$\alpha = \frac{glm(M+m)\sin(\theta) - lm(-bv+lm\omega^2\sin(\theta)+u)\cos(\theta)}{-l^2m^2\cos(\theta)^2 + (I+l^2m)(M+m)} \quad \text{(S11)}$$

$$E = glm\cos(\theta) + lm\omega v\cos(\theta) + \frac{\omega^2(I+l^2m)}{2} + \frac{v^2(M+m)}{2} \quad \text{(S12)}$$

In the acceleration equations, replace u by u + fpush when applying the external disturbance. With $J = (1+\rho)ml^2$, $\Delta = ml^2[M + (M+m)\rho + m\sin^2\theta] > 0$ for M,m,l > 0 and ρ ≥ 0. The energy derivative is $\dot{E} = (u + f_{push})v - bv^2$. This balance checks the sign of gravity, friction and input coupling. The rod shares l with the point mass and has physical length 2l; equating physical pole length would define a different comparison.

2 Upright linearization and discretization

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -\frac{b(I+l^2m)}{-l^2m^2+(I+l^2m)(M+m)} & -\frac{gl^2m^2}{-l^2m^2+(I+l^2m)(M+m)} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & \frac{blm}{-l^2m^2+(I+l^2m)(M+m)} & \frac{glm(M+m)}{-l^2m^2+(I+l^2m)(M+m)} & 0 \end{bmatrix} \quad \text{(S13)}$$

$$\begin{bmatrix} 0 \\ \frac{I+l^2m}{-l^2m^2+(I+l^2m)(M+m)} \\ 0 \\ -\frac{lm}{-l^2m^2+(I+l^2m)(M+m)} \end{bmatrix} \quad \text{(S14)}$$

Linearization is about z = 0, u = 0. The input sign follows θ positive toward positive x. The controller convention u = −Kz therefore gives negative entries in K for these coordinates. Independent central differences of the nonlinear RHS check both Jacobians. Exact zero-order-hold discretization follows from the block exponential exp([[A,B],[0,0]]Ts); the upper blocks are F and G.

$$F = e^{AT_s}, G = \int_0^{T_s} e^{A\tau} B \mathrm{d}\tau \quad \text{(S15)}$$

Point nominal plant

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -0.1 & -0.981 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0.2 & 21.582 & 0 \end{bmatrix} \quad \text{(S16)}$$

$$B = \begin{bmatrix} 0 \\ 1 \\ 0 \\ -2 \end{bmatrix} \quad \text{(S17)}$$

Rod nominal plant

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -0.097561 & -0.7178 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0.14634 & 15.792 & 0 \end{bmatrix} \quad \text{(S18)}$$

$$B = \begin{bmatrix} 0 \\ 0.97561 \\ 0 \\ -1.4634 \end{bmatrix} \quad \text{(S19)}$$

Plant integration uses RK4 at 1 ms, whereas F and G are exact linear sample transitions at Ts = 10 ms. Step-halving verification compares the same held-control schedule, including push boundaries, at 1 ms and 0.5 ms. The integration tolerance is fixed before examining outcomes.

3 Continuous and discrete quadratic regulators

The nominal continuous cost is $\int(z^TQz + Ru^2)dt$. Bryson limits 1 m, 1 m/s, 0.2 rad, 1 rad/s and 10 N give Q = diag(1,1,25,1), R = 0.01. Completing the Hamilton–Jacobi quadratic identity gives the stabilizing CARE and $u = -R^{-1}B^TPz$.

$$A^TP + PA - PBR^{-1}B^TP + Q = 0 \quad \text{(S20)}$$

$$K = R^{-1}B^TP \quad \text{(S21)}$$

$$u = -K\hat{z} \quad \text{(S22)}$$

$$P = F^TPF - F^TPG(RT_s + G^TPG)^{-1}G^TPF + QT_s \quad \text{(S23)}$$

$$K_d = (RT_s + G^TPG)^{-1}G^TPF \quad \text{(S24)}$$

The discrete stage cost is $Ts(z^TQz + Ru^2)$. The displayed DARE follows by substituting znext = Fz + Gu into the Bellman quadratic and minimizing with respect to u. An exact integral cost would introduce different weights and a state-input cross term; this study consistently uses the declared sampled-stage cost.

Point regulator gains

$$K = [-10 \quad -17.42 \quad -108.88 \quad -23.789] \quad \text{(S25)}$$

$$\mathrm{Kd} = [-8.6023 \quad -15.045 \quad -95.956 \quad -20.967] \quad \text{(S26)}$$

Rod regulator gains

$$K = [-10 \quad -17.865 \quad -116.54 \quad -29.008] \quad \text{(S27)}$$

$$\mathrm{Kd} = [-8.8251 \quad -15.825 \quad -104.93 \quad -26.145] \quad \text{(S28)}$$

4 Finite differences and local filter aware gain redesign

$$H_v(z) = \frac{(1-\alpha)(1-z^{-1})}{T_s(1-\alpha z^{-1})} \quad \text{(S29)}$$

$$h_{k+1} = \alpha h_k + (1-\alpha) y_k \quad \text{(S30)}$$

$$\hat{v}_k = \frac{1-\alpha}{T_s}(y_k - h_k) \quad \text{(S31)}$$

$$K_y = K_p + \frac{1-\alpha}{T_s} K_v \quad \text{(S32)}$$

$$K_h = -\frac{1-\alpha}{T_s} K_v \quad \text{(S33)}$$

Here h stores the exponentially filtered previous measurement and h0 = y0; Kp and Kv select the position and velocity columns from K. Then u = −Ky y − Kh h. In the augmented state s = [z;h], Ag = [[F,0],[(1−α)C,αI]], Bg = [G;0], Cy = diag(C,I), Ko = [Ky,Kh] and Acl = Ag − BgKoCy. This realization includes the derivative filter in the loop. A fixed PD cascade also gives four gains K = −[ka kx, ka kv, ka, kw], so its linear implementation is in this same family.

Set Qg = diag(QTs,0), Qcl = Qg + (KoCy)^{T}R(KoCy)Ts and Bn = [−GKy;(1−α)I]. For a Schur-stable candidate, P sums the initial transient quadratic cost, and S is the stationary state covariance driven by independent measurement noise:

$$P = {A_{\mathrm{cl}}}^{T} P A_{\mathrm{cl}} + Q_{\mathrm{cl}} \quad \text{(S34)}$$

$$S = A_{\mathrm{cl}} S {A_{\mathrm{cl}}}^{T} + B_n R_m {B_n}^{T} \quad \text{(S35)}$$

$$J_{\mathrm{sur}} = {s_0}^{T} P s_0 + N\mathrm{tr}(Q_{\mathrm{cl}} S) + N T_s R {K_y} R_m {K_y}^{T} \quad \text{(S36)}$$

N = 1000 samples represents ten seconds. The final term accounts for direct measurement-noise feedthrough to command effort. The initial augmented state is s0 = [0,0,0.1,0,0,0.1]^{T}. Gain signs are preserved using K = Kbase diag(exp ℓi), with ℓi ∈ [−2.3,2.3]. A fixed-seed set of 120 random starts plus zero is ranked, and the best three seed Powell local searches. Unstable candidates receive a spectral-radius penalty and cannot be selected as stable designs. The surrogate omits saturation, delay, parameter errors and the unknown push; its purpose is a transparent filter-aware baseline rather than a global robust optimum.

All nine α values are evaluated with fixed K and with this redesign, where stable. Nonlinear development tasks select α afterward without using held-out outcomes. The JSON gain audit reports coefficients, objective values, local-search status and stability. The following tables give the final primary configurations; repeated settings retain separate diagnostic labels.

Point primary gains

Feedback	α	σ	Kx	Kv	Kθ	Kω
PD reference	0.80	1	-6	-12	-60	-12
Velocity oracle	0.80	1	-10	-17.42	-108.88	-23.789
FD reference	0.80	1	-10	-17.42	-108.88	-23.789
FD selected	0.00	1	-10	-17.42	-108.88	-23.789
FD redesigned	0.40	1	-1.5909	-1.7465	-34.715	-3.6367
cKF legacy	0.80	1	-10	-17.42	-108.88	-23.789
cKF units fixed	0.80	1	-10	-17.42	-108.88	-23.789
cKF causal	0.80	1	-10	-17.42	-108.88	-23.789
cKF selected	0.80	10	-10	-17.42	-108.88	-23.789
dKF base	0.80	1	-8.6023	-15.045	-95.956	-20.967
dKF selected	0.80	10	-8.6023	-15.045	-95.956	-20.967
Predictor base	0.80	1	-8.6023	-15.045	-95.956	-20.967
Predictor selected	0.80	10	-8.6023	-15.045	-95.956	-20.967
dKF continuous K	0.80	1	-10	-17.42	-108.88	-23.789

Rod primary gains

Feedback	α	σ	Kx	Kv	Kθ	Kω
PD reference	0.80	1	-6	-12	-60	-18
Velocity oracle	0.80	1	-10	-17.865	-116.54	-29.008
FD reference	0.80	1	-10	-17.865	-116.54	-29.008
FD selected	0.10	1	-10	-17.865	-116.54	-29.008
FD redesigned	0.60	1	-1.7751	-1.7912	-40.058	-4.7128
cKF legacy	0.80	1	-10	-17.865	-116.54	-29.008
cKF units fixed	0.80	1	-10	-17.865	-116.54	-29.008
cKF causal	0.80	1	-10	-17.865	-116.54	-29.008
cKF selected	0.80	1	-10	-17.865	-116.54	-29.008
dKF base	0.80	1	-8.8251	-15.825	-104.93	-26.145
dKF selected	0.80	100	-8.8251	-15.825	-104.93	-26.145
Predictor base	0.80	1	-8.8251	-15.825	-104.93	-26.145
Predictor selected	0.80	100	-8.8251	-15.825	-104.93	-26.145
dKF continuous K	0.80	1	-10	-17.865	-116.54	-29.008

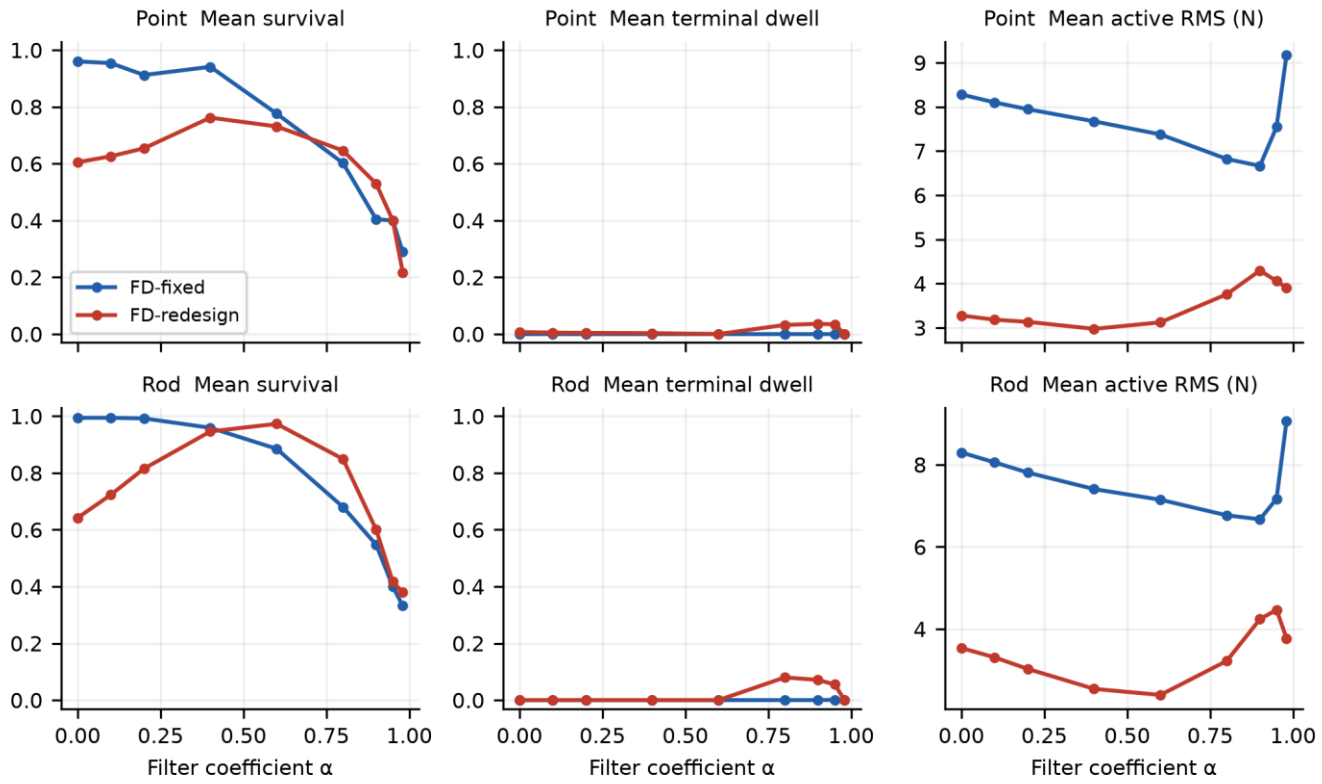


Fig. S1. Full development α sweep: mean survival, terminal dwell and active-time RMS across the five declared tasks, for fixed and filter-aware redesigned gains. Each setting uses 200 paired trials per task.

5 Covariance units continuous observation and sampled filtering

Per-sample covariance Rm has units of squared measurement, whereas continuous white-noise spectral density Vc has units squared measurement times seconds. Averaging a white-noise measurement over Ts gives covariance Vc/Ts; matching this to Rm gives the stated conversion. Independent point samples do not define a finite continuous white-noise variance without such a bandwidth/averaging convention, so this is an explicit equivalent-noise convention rather than an assertion of exact continuous/discrete equivalence.

$$V_c = R_m T_s \quad \text{(S37)}$$

$$A\Sigma + \Sigma A^T - \Sigma C^T V^{-1} C\Sigma + W = 0 \quad \text{(S38)}$$

$$H = \Sigma C^T V^{-1} \quad \text{(S39)}$$

$$H(\sigma W, V) = H(W, \frac{V}{\sigma}) \quad \text{(S40)}$$

The continuous dual CARE is $A\Sigma + \Sigma A^T - \Sigma C^T V^{-1} C\Sigma + \sigma W = 0$. Rescaling both σW and V by the same positive constant rescales Σ but leaves H invariant. Thus correcting V from Rm to RmTs is equivalent at Ts = 0.01 s to multiplying W by 100 in the legacy design. The numerical unit-ratio identity is checked independently for both plants. At extreme ratios, covariance and time rescaling condition the CARE solve; accepted solutions must have relative residual below 10^{-6} and stable observer poles.

For a causal continuous observer, the exact held transition is generated by the upper blocks of exp([[A−HC,B,H],[0,0,0],[0,0,0]]Ts). Its propagation over [tk−1,tk) uses u at k−1 and y at k−1, then u at k = −Kẑk. The legacy-current-sample diagnostic uses y at k in that same preceding-interval propagation. This alternative has a different sampled transfer and cannot be called a causal continuous implementation. The old-unit and corrected-unit legacy diagnostics share the exact same update, isolating units alone.

$$Q_d = \int_0^{T_s} e^{A\tau} W e^{A^T \tau} \mathrm{d}\tau \quad \text{(S41)}$$

$$L = \Sigma C^T (C\Sigma C^T + R_m)^{-1} \quad \text{(S42)}$$

$$\Sigma_+ = (I - LC)\Sigma_- \quad \text{(S43)}$$

$$\Sigma = F\Sigma F^T + Q_d - F\Sigma C^T (C\Sigma C^T + R_m)^{-1} C\Sigma F^T \quad \text{(S44)}$$

W = diag(10^{-6},0.04,10^{-6},0.04) is a continuous design covariance; Qd is its exact integral [2]. Rm = diag(0.005^2,0.002^2) is used directly in the discrete innovation covariance. Prediction uses the actual clipped command. The steady-state prior covariance is computed from the dual DARE; the gain updates the current prior with yk. At zero delay, the post-update compensator has Ac = (I−LC)(F−GKd) and transfer Kd(I−Ac z^{-1})$^{-1}$L. Discrete and continuous regulator gains are not silently interchanged.

Each observer family is swept over σ = 10^j, j = −6,…,8. Because plant process noise is not drawn from W, neither observer is asserted statistically optimal for the nonlinear uncertain plant. The sweep is a bandwidth calibration study. Survival maxima are task dependent; dwell and effort need not peak at the same ratio. A finite tested maximum is distinguished from an identified unique continuous optimum. At σ = 10^8, the discrete gain change from the preceding decade is below 10^{-4} in relative norm, documenting convergence toward the high-bandwidth limit.

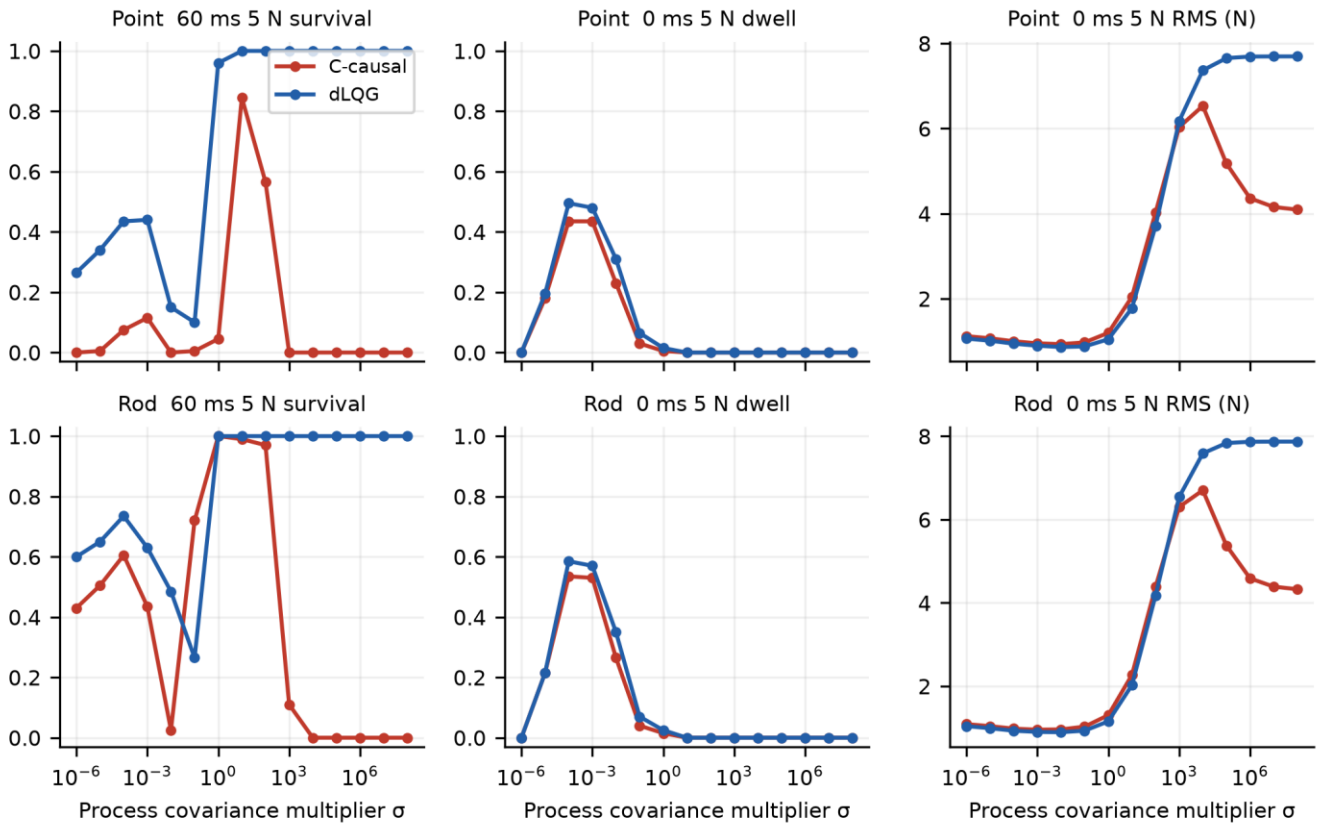


Fig. S2. Full covariance-ratio trade-off: 60 ms/5 N survival and zero-delay/5 N terminal dwell and active RMS, across both models and both corrected observer families. The full tested multiplier range extends fourteen orders of magnitude.

Point base estimator

$$H = \begin{bmatrix} 28.254 & 0.34068 \\ 397.17 & 5.0676 \\ 0.05451 & 45.483 \\ 3.2086 & 1021.9 \end{bmatrix} \quad \text{(S45)}$$

$$L = \begin{bmatrix} 0.24613 & 0.0023517 \\ 3.4483 & 0.034815 \\ 0.00037628 & 0.36543 \\ 0.02563 & 8.1431 \end{bmatrix} \quad \text{(S46)}$$

$$\mathrm{Qd} = \begin{bmatrix} 2.3323 \times 10^{-8} & 1.998 \times 10^{-6} & 9.7001 \times 10^{-12} & 2.6489 \times 10^{-9} \\ 1.998 \times 10^{-6} & 0.0003996 & 1.234 \times 10^{-9} & 3.9312 \times 10^{-7} \\ 9.7001 \times 10^{-12} & 1.234 \times 10^{-9} & 2.3346 \times 10^{-8} & 2.0025 \times 10^{-6} \\ 2.6489 \times 10^{-9} & 3.9312 \times 10^{-7} & 2.0025 \times 10^{-6} & 0.00040029 \end{bmatrix} \quad \text{(S47)}$$

Rod base estimator

$$H = \begin{bmatrix} 28.257 & 0.24971 \\ 397.24 & 3.7184 \\ 0.039954 & 45.353 \\ 2.3461 & 1015.9 \end{bmatrix} \quad \text{(S48)}$$

$$L = \begin{bmatrix} 0.24616 & 0.0017249 \\ 3.4489 & 0.025568 \\ 0.00027598 & 0.3646 \\ 0.018744 & 8.1004 \end{bmatrix} \quad \text{(S49)}$$

$$\mathrm{Qd} = \begin{bmatrix} 2.3324\times10^{-8} & 1.998\times10^{-6} & 7.0976\times10^{-12} & 1.9381\times10^{-9} \\ 1.998\times10^{-6} & 0.00039961 & 9.0295\times10^{-10} & 2.8765\times10^{-7} \\ 7.0976\times10^{-12} & 9.0295\times10^{-10} & 2.3343\times10^{-8} & 2.0018\times10^{-6} \\ 1.9381\times10^{-9} & 2.8765\times10^{-7} & 2.0018\times10^{-6} & 0.00040021 \end{bmatrix} \quad \text{(S50)}$$

6 Timestamp aware prediction with held command history

Write $\tau = qT_s + r$, $0 \le r < T_s$, $F_s = \exp(As)$, $G_s = \int_0^s \exp(Aa)B\,da$, and $Q_s = \int_0^s \exp(Aa)\sigma W \exp(A^T a)\,da$. The incoming packet at t_k observes timestamp $s_k = t_k - \tau$; write $s = s_k$ in the forecast below. Stale timestamps remain exactly T_s apart, but their propagation interval may straddle two command holds. From the previous stale posterior to the new stale prior,

$$\hat{z}_{s_k|k-1} = F\hat{z}_{s_{k-1}|k-1} + F_{T_s-r}G_r u_{k-q-2} + G_{T_s-r}u_{k-q-1} \quad \text{(S51)}$$

The stale covariance propagates with F and Qd. Innovation $y_k - C\hat{z}_{s^-}$ and gain $P_{s^-}C^T(CP_{s^-}C^T+R_m)^{-1}$ update the stale state. At zero delay, the steady-state gain is used exactly, ensuring equality with the ordinary discrete filter. At nonzero delay, covariance is propagated from the first received timestamp. After that innovation, the present-time mean is

$$\hat{z}_k = F_\tau \hat{z}_{s|k} + \sum_{h=0}^{q-1} F^h G u_{k-1-h} + F^q G_r u_{k-q-1} \quad \text{(S52)}$$

$$\hat{z}_k = F^q \hat{z}_{k-q|k} + \sum_{h=0}^{q-1} F^h G u_{k-1-h} \quad \text{(S53)}$$

The first partial interval lasts r with command u at $k-q-1$; the remaining q complete intervals use the chronological stored commands. The sum is written backward from the most recent input to avoid an indexing ambiguity. Before any valid packet arrives, the current mean propagates from the common known initial mean. The first stale update occurs at a timestamp between zero and T_s; its transition uses only the initial held command. The predictor never knows the external push, the trial-specific parameter perturbations or future commands.

For an exact linear model with known inputs, no unknown disturbance and consistent initial mean, integrating the stale posterior and command history recovers the current state. Hence deterministic delay disappears from regulation in that ideal limit. With stochastic process noise, prediction uncertainty accumulates during τ; with an unknown push or parameter errors, current-time extrapolation is imperfect. No infinite nonlinear delay tolerance is claimed. This predictor is a standard baseline from delayed-measurement filtering [3], not an algorithmic contribution.

7 Exact sampled measurement delay and local stability

For a fixed compensator $\xi_{next} = A_c\xi + B_c y$ and command $u = -C_c\xi - D_c y$, reverse the nominal plant transition over r within the previous held-input segment. With $F_r = \exp(Ar)$, the fractional delayed output is

$$y_k = CF_r^{-1}z_{k-q} - CF_r^{-1}G_r u_{k-q-1} \quad \text{(S54)}$$

Here $F_r^{-1} = \exp(-Ar)$; for $r = 0$ the input correction is zero. The augmented state contains z at k through $k-q$, commands at $k-1$ through $k-q-1$, and controller state ξ_k. The output expression determines y, then u; the plant advances using F and G, histories shift, and the compensator advances. The resulting finite matrix gives the exact nominal closed-loop spectral radius at that τ.

Zero-delay augmented radii are checked against an independently assembled python-control feedback interconnection. All gain crossovers are computed from state-space frequency responses below Nyquist. The smallest positive φ_m/ω_{gc} estimates the first pure-delay crossing. Exact fractional-delay radii are evaluated on a 10 ms locator and bisected to 1 ms. The reported linear boundary is the bracket midpoint; half the bracket width is numerical resolution, not statistical uncertainty. Neither method certifies nonlinear survival under saturation or uncertainty.

$$\tau_{PM} = \min_i \frac{\varphi_{mi}}{\omega_{gci}} \quad \text{(S55)}$$

Model and loop	Phase deg	ωgc rad/s	PM delay ms	Lift delay ms
point PD reference	27.3	13.66	34.9	34.5
point Velocity oracle	68.6	29.96	40.0	39.5
point FD reference	22.2	25.37	15.3	15.5
point FD selected	60.3	30.60	34.4	34.5
point FD redesigned	26.5	7.64	60.5	60.5
point cKF legacy	27.8	8.65	56.2	56.5
point cKF units fixed	43.3	17.30	43.7	43.5

Model and loop	Phase deg	ωgc rad/s	PM delay ms	Lift delay ms
point cKF causal	33.4	17.30	33.7	33.5
point cKF selected	41.5	21.06	34.4	34.5
point dKF base	40.6	17.20	41.2	41.5
point dKF selected	51.5	21.25	42.3	42.5
point dKF continuous K	40.1	18.28	38.3	38.5
rod PD reference	35.4	14.34	43.1	43.5
rod Velocity oracle	68.5	24.87	48.1	48.5
rod FD reference	25.9	22.27	20.3	20.5
rod FD selected	60.3	25.50	41.3	41.5
rod FD redesigned	27.4	7.36	65.0	64.5
rod cKF legacy	27.9	8.49	57.3	57.5
rod cKF units fixed	44.2	16.34	47.2	47.5
rod cKF causal	34.8	16.34	37.2	37.5
rod cKF selected	34.8	16.34	37.2	37.5
rod dKF base	41.9	16.31	44.9	44.5
rod dKF selected	60.6	20.58	51.3	51.5
rod dKF continuous K	41.2	17.25	41.7	41.5

8 Paired experiments boundary estimation and censoring

Development uses seed 20261121 and 200 trials per each of five tasks. The coarse delay locator uses seed 20261124 and 100 trials per cell. Held-out boundary and fixed reference cells use seed 20261122 and 500 trials per cell. Four independent uniform parameter draws are generated once per trial and shared across all compared cases; at each control instant, a single two-channel measurement-noise draw per trial is likewise shared. This construction is independent of batching and case order, an invariant verified numerically. The pair represents a common realization, not independent observations across delays.

$$D_{50} = \inf\{\tau \geq 0 : p_T(\tau) < \tfrac{1}{2}\} \quad \text{(S56)}$$

$$\hat{D}_{50} = \tau_a + (\tau_b - \tau_a)\frac{\hat{p}_a - .5}{\hat{p}_a - \hat{p}_b} \quad \text{(S57)}$$

The held-out bracket is verified before bisection and expanded if necessary, with an administrative search ceiling of 1200 ms. The actual evaluated maxima are 300 ms for point and 600 ms for rod; no reported crossing is right-censored. A ±6 ms local integer grid around the final crossing checks local recrossings; some values were already queried and are reused. Coarse locator and held-out tables retain every evaluated delay. If survival is nonmonotone, the estimand is the first observed downward crossing, and observed upward recrossings are counted rather than forced into a monotone fit. An unobserved narrow dip between coarse locator points cannot be excluded.

Bootstrap seed 20261123 resamples 500 trial identities with replacement, recomputes all evaluated survival means, and interpolates the first crossing; 2,000 resamples yield 2.5th and 97.5th percentiles. Reported bounds round the bootstrap endpoints outward to integer milliseconds and include the observed grid crossing bracket; the unrounded interpolation intervals are also supplied. The intervals condition on the sampled adaptive delay grid and fixed selected design. They do not claim coverage of a repeated adaptive sampling/tuning procedure. A bootstrap realization that never crosses is excluded from finite quantiles and its occurrence is reported through the valid-resample count; a zero-delay or upper-limit censoring status takes precedence over any finite interval.

$$\frac{\hat{p} + \frac{\zeta^2}{2n} \pm \zeta\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{\zeta^2}{4n^2}}}{1 + \frac{\zeta^2}{n}} \quad \text{(S58)}$$

$$\Delta\mathrm{p} = \frac{1}{n}\sum_{i=1}^{n}(Y_{\mathrm{Bi}} - Y_{\mathrm{Ai}}) \quad \text{(S59)}$$

Pointwise Wilson intervals [4] describe each fixed survival cell. For the 108 declared fixed-cell paired interventions, Δp is the mean within-trial success difference b − a. Positive values favor b. Seed 20261128 provides 2,000 paired bootstrap

resamples. Exact McNemar tests use only the improved and worsened discordant pairs [5]; Holm correction is applied across all 108 p-values [6]. Adaptive D50 comparisons are not added to that family after inspecting outcomes. These tests quantify the specified interventions and are not a ranking of all controller theory.

Active RMS averages each trajectory's squared clipped command over timesteps before its first failure, including the threshold-crossing step; full RMS uses all ten seconds, including the held command until the first control update after failure and zeros thereafter. Reported aggregate RMS is the mean of per-trial RMS values, not the square root of pooled effort. Terminal dwell is a separate binary event and requires survival. The zero-action tail after failure therefore cannot create a favorable active-effort result.

9 Full boundary results and verification

Point held out boundaries

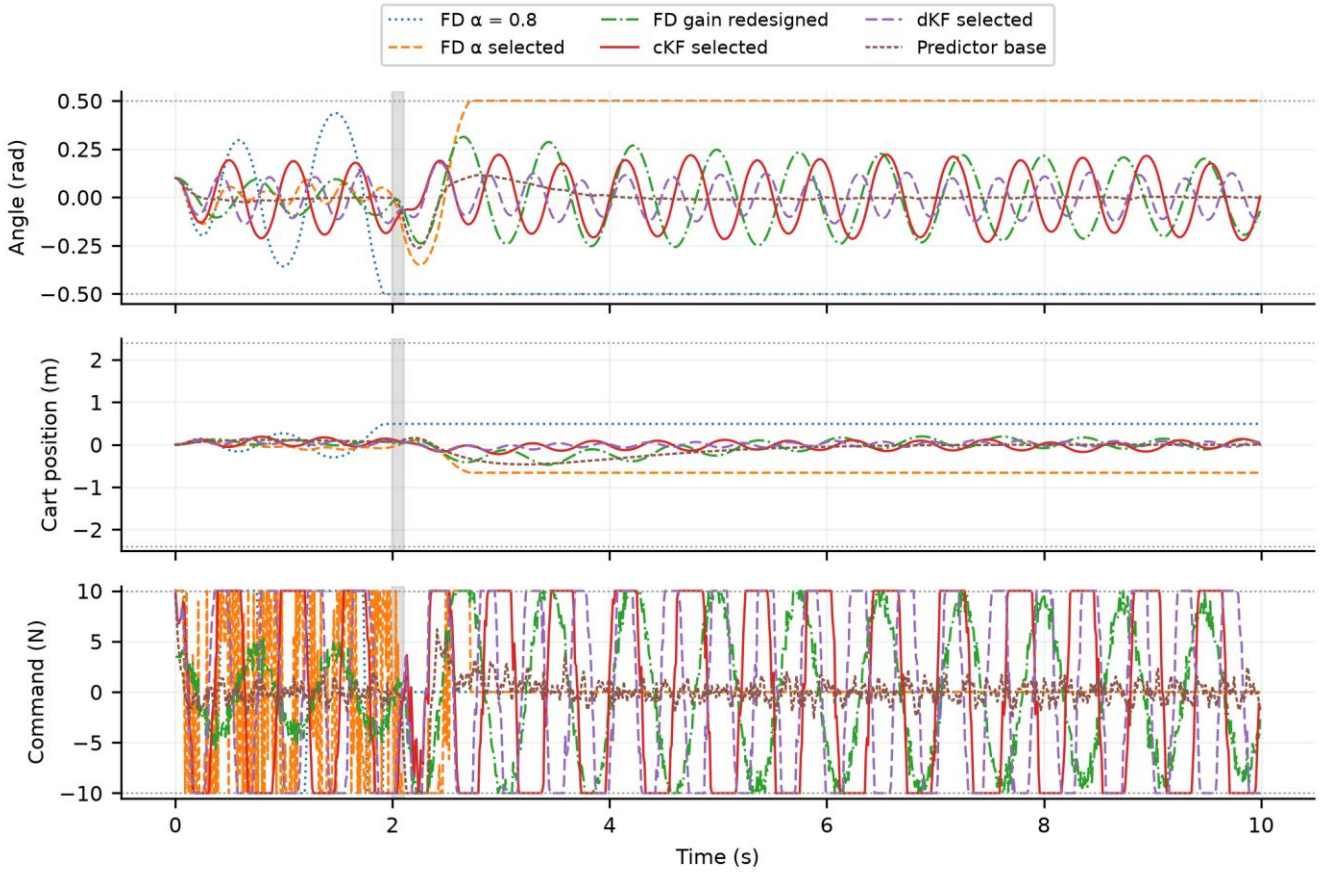


Fig. S3. One paired illustrative point realization at 60 ms/10 N, seed 20261125. Gray shading marks the unknown push and dotted horizontal lines mark failure or actuator limits. Failed states are frozen and commands are zeroed at the next control update. These trajectories illustrate behavior and do not estimate ensemble survival.

Feedback	Push N	D50 ms	95% interval ms	Upward recrossings
PD reference	5	38.9	38.0 to 40.0	0
PD reference	10	36.3	32.0 to 37.0	0
Velocity oracle	5	77.4	77.0 to 78.0	0
Velocity oracle	10	56.6	56.0 to 57.0	2
FD reference	5	51.9	51.0 to 53.0	0
FD reference	10	31.7	31.0 to 32.0	2
FD selected	5	76.7	76.0 to 78.0	0
FD selected	10	69.0	67.0 to 70.0	0
FD redesigned	5	60.7	60.0 to 61.0	0
FD redesigned	10	55.1	54.0 to 56.0	0

Feedback	Push N	D50 ms	95% interval ms	Upward recrossings
cKF legacy	5	59.2	58.0 to 61.0	0
cKF legacy	10	47.8	47.0 to 49.0	0
cKF units fixed	5	59.6	59.0 to 60.0	1
cKF units fixed	10	46.7	46.0 to 47.0	1
cKF causal	5	49.6	49.0 to 50.0	1
cKF causal	10	36.7	36.0 to 37.0	1
cKF selected	5	55.5	55.0 to 56.0	1
cKF selected	10	38.6	38.0 to 39.0	3
dKF base	5	66.3	66.0 to 67.0	0
dKF base	10	44.1	43.0 to 45.0	1
dKF selected	5	64.9	64.0 to 66.0	1
dKF selected	10	47.3	46.0 to 48.0	3
Predictor base	5	252.2	250.0 to 254.0	0
Predictor base	10	125.5	123.0 to 128.0	0
Predictor selected	5	262.2	260.0 to 264.0	0
Predictor selected	10	134.5	132.0 to 137.0	0
dKF continuous K	5	65.9	58.0 to 67.0	0
dKF continuous K	10	44.1	43.0 to 45.0	1

Rod held out boundaries

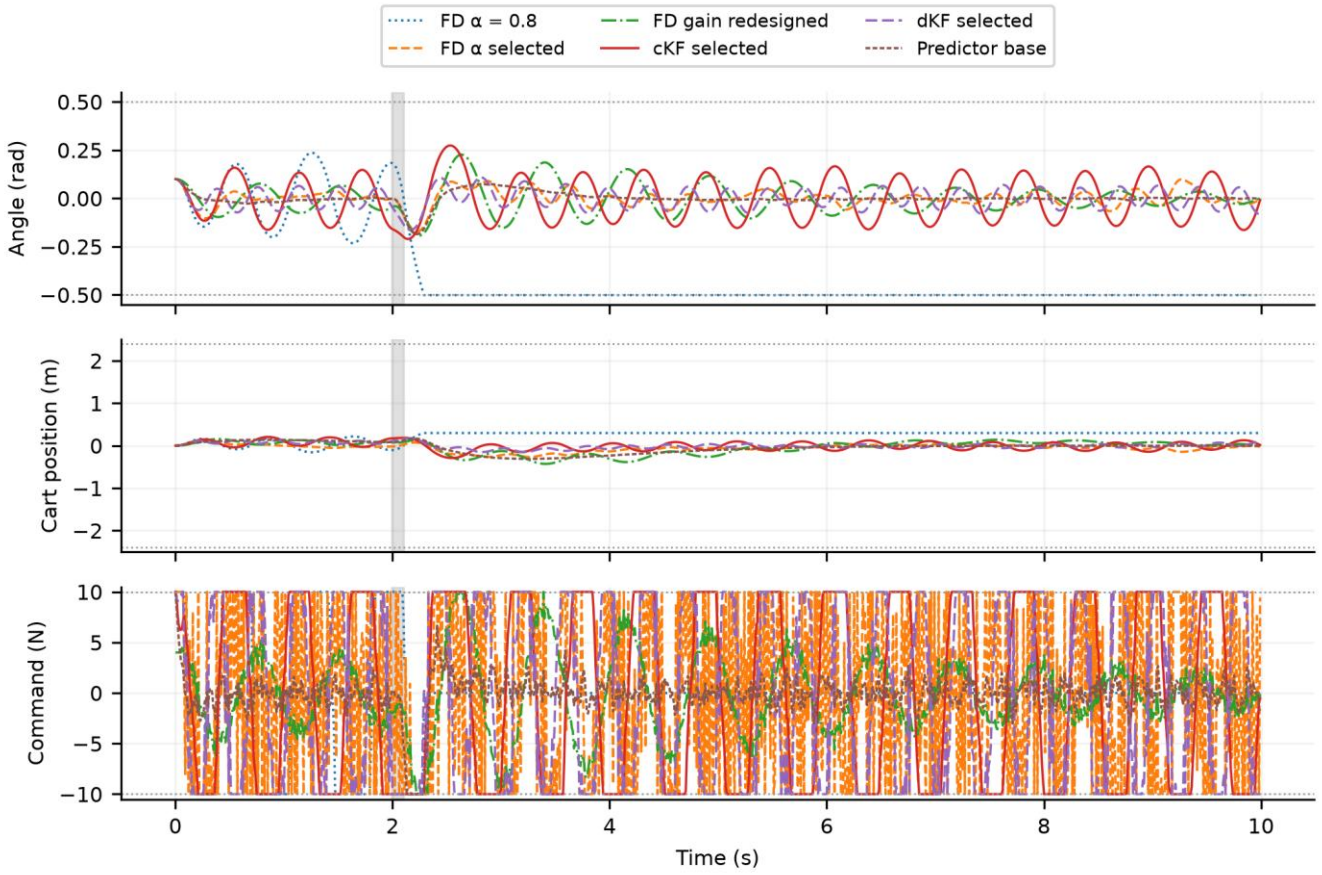


Fig. S4. One paired illustrative rod realization at 60 ms/10 N, seed 20261125. Gray shading marks the unknown push and dotted horizontal lines mark failure or actuator limits. Failed states are frozen and commands are zeroed at the next control update. These trajectories illustrate behavior and do not estimate ensemble survival.

Feedback	Push N	D50 ms	95% interval ms	Upward recrossings
PD reference	5	63.3	63.0 to 64.0	0
PD reference	10	49.0	48.0 to 50.0	1
Velocity oracle	5	82.6	82.0 to 83.0	0
Velocity oracle	10	66.8	66.0 to 74.0	1
FD reference	5	57.3	56.0 to 58.0	0
FD reference	10	54.1	53.0 to 55.0	0
FD selected	5	84.1	83.0 to 85.0	0
FD selected	10	79.2	75.0 to 81.0	0
FD redesigned	5	66.5	66.0 to 67.0	0
FD redesigned	10	63.0	62.0 to 64.0	0
cKF legacy	5	63.4	62.0 to 65.0	0
cKF legacy	10	58.2	57.0 to 59.0	0
cKF units fixed	5	73.2	73.0 to 74.0	0
cKF units fixed	10	59.0	58.0 to 60.0	0
cKF causal	5	63.1	63.0 to 64.0	0
cKF causal	10	48.9	48.0 to 50.0	1
cKF selected	5	63.1	63.0 to 64.0	0
cKF selected	10	48.9	48.0 to 50.0	1
dKF base	5	71.5	71.0 to 72.0	0
dKF base	10	56.9	56.0 to 58.0	0
dKF selected	5	81.2	80.0 to 82.0	0
dKF selected	10	70.5	70.0 to 71.0	1
Predictor base	5	311.4	308.0 to 317.0	0
Predictor base	10	173.3	171.0 to 175.0	0
Predictor selected	5	314.0	309.0 to 319.0	0
Predictor selected	10	181.1	178.0 to 183.0	0
dKF continuous K	5	70.9	70.0 to 71.0	0
dKF continuous K	10	56.8	56.0 to 57.0	0

The revision validation report contains 158 recorded checks and diagnostics. It includes nonlinear Jacobians, integration step halving, covariance-unit gain identity, predictor equality at zero delay, exact known-model fractional-delay recovery, random held-command convolution, nominal closed-loop radii, case-order/batch invariance and interpolation. Riccati residual and pole checks are enforced for every design. Margin checks independently compare sampled interconnections. Video checks read the first and final frames and verify 240 frames at 24 fps.

The boundary campaign evaluates 596,000 nonlinear trajectories, including its independent locators. Development adds 96,000 trajectories across 18 derivative-filter configurations and 30 covariance settings per plant, five tasks and 200 trials. These counts exclude implementation checks and the 168 single-case video trajectories. The trial census and all queried cells are supplied to make the accounting auditable.

Randomized push phase sensitivity

The phase check uses 500 matched trials per regime and 269 model-controller-push-delay cells. For each original first-crossing bracket [a,b], the queried delays are max(0,a−3), a, b, b+3 and 60 ms, with duplicates removed. Phase seed 20261106 draws independent uniform integers 0,…,9; the same phase draw is shared across configurations and models for each trial. A pulse starts at 2000 + phase integration steps and ends 100 steps later. Plant and measurement seed 20261122 is unchanged. All 269 fixed-phase reruns reproduce the archived binary survival outcomes exactly; the phase extension evaluates 269,000 trajectories.

Seed 20261107 supplies 2,000 trial-paired resamples for pointwise 95% intervals of randomized-minus-fixed survival. These exploratory intervals are not multiplicity-adjusted. Seven evaluated cells change sides of 50%; the largest observed absolute differences are 2.8 points at 60 ms and 5.4 points at local anchors. No new phase-averaged D50, global phase robustness or arbitrary-time disturbance tolerance is inferred. Raw trial outcomes, assignments and aggregate tables accompany this supplement.

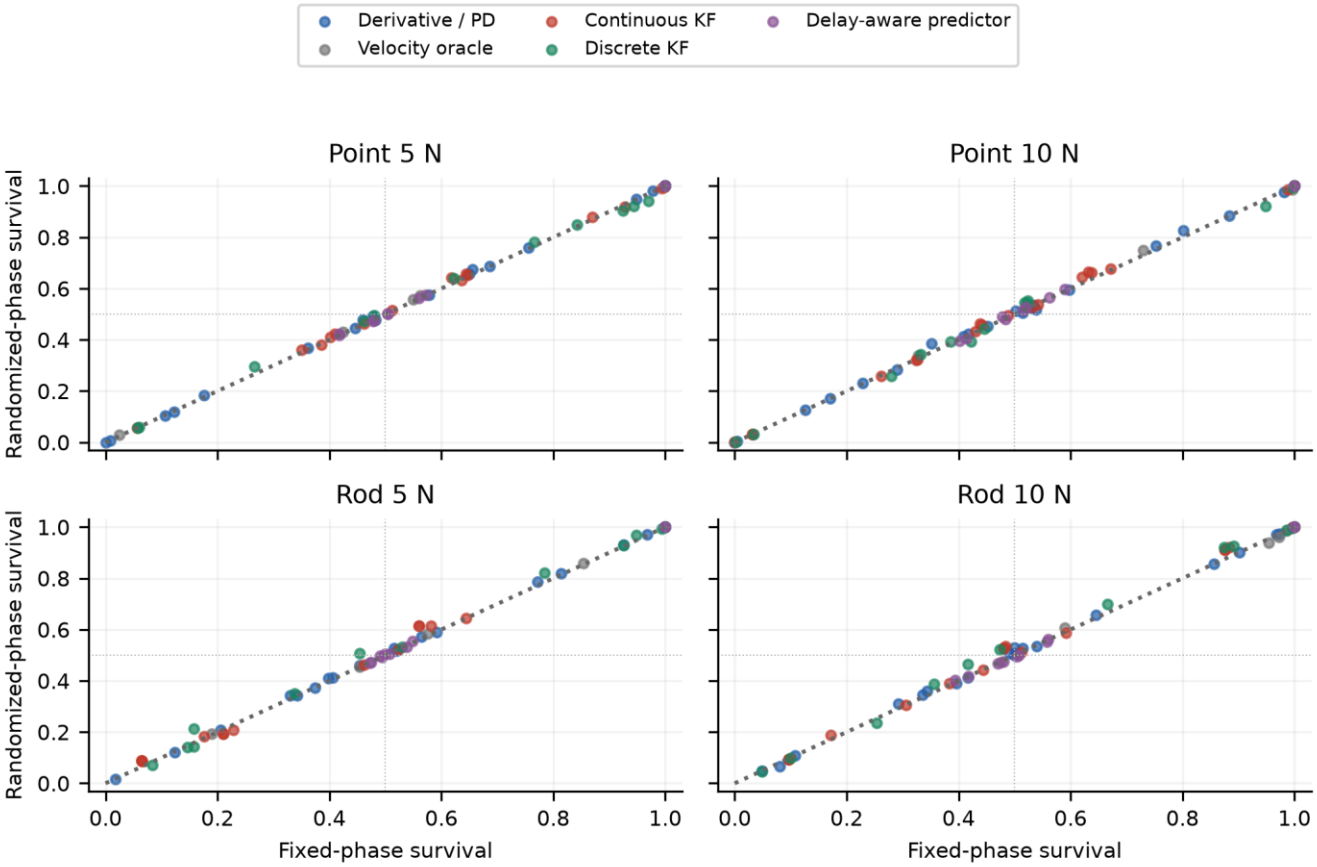


Fig. S5. Randomized-minus-fixed push-phase sensitivity across all fourteen configurations, both push magnitudes and the queried local-crossing/60 ms cells. Each point compares 500 trial-paired realizations per regime; the diagonal is equality. The push starts at 2 s plus an independent uniform integer 0,…,9 ms and lasts 100 ms. This local sensitivity check does not re-estimate a full phase-averaged D50 curve.

10 Scope and reproducibility

Run development, boundary estimation, margins, audits, figures and videos with the supplied sources and recorded seeds. The document formatter uses editable Office Math and the supplied mechanical MathML. Regenerating Word/PDF on this host uses Microsoft Word 16; scientific computation does not require Word. No hardware data, identified sensor spectrum or verified journal acceptance is implied. Generalization requires additional plants, physical experiments, delay jitter/loss and broader tuning objectives.

REFERENCES

[1] K. J. Åström and R. M. Murray, Feedback Systems: An Introduction for Scientists and Engineers. Princeton University Press, 2008. https://authors.library.caltech.edu/records/yzs24-xsx88

[2] C. F. Van Loan, “Computing integrals involving the matrix exponential,” IEEE Trans. Autom. Control, vol. 23, no. 3, pp. 395–404, 1978, doi: 10.1109/TAC.1978.1101743.

[3] T. D. Larsen, N. A. Andersen, O. Ravn, and N. K. Poulsen, “Incorporation of time delayed measurements in a discrete-time Kalman filter,” Proc. 37th IEEE Conf. Decision and Control, vol. 4, pp. 3972–3977, 1998, doi: 10.1109/CDC.1998.761918.

[4] E. B. Wilson, “Probable inference, the law of succession, and statistical inference,” J. Amer. Stat. Assoc., vol. 22, no. 158, pp. 209–212, 1927, doi: 10.1080/01621459.1927.10502953.

[5] Q. McNemar, “Note on the sampling error of the difference between correlated proportions or percentages,” Psychometrika, vol. 12, no. 2, pp. 153–157, 1947. https://pubmed.ncbi.nlm.nih.gov/20254758/

[6] S. Holm, “A simple sequentially rejective multiple test procedure,” Scandinavian J. Stat., vol. 6, no. 2, pp. 65–70, 1979. https://sci2s.ugr.es/keel/pdf/specific/articulo/0052_001.pdf